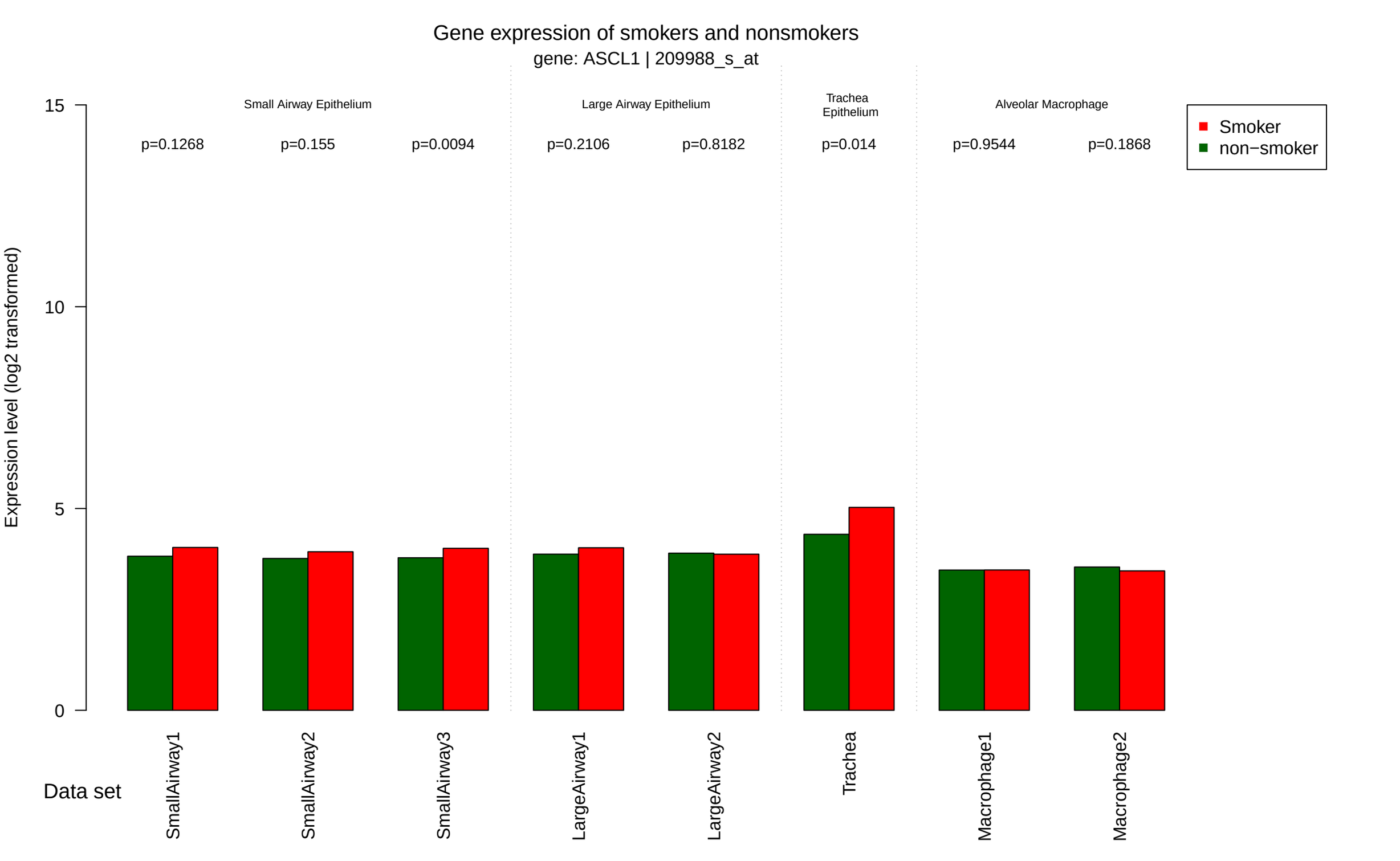
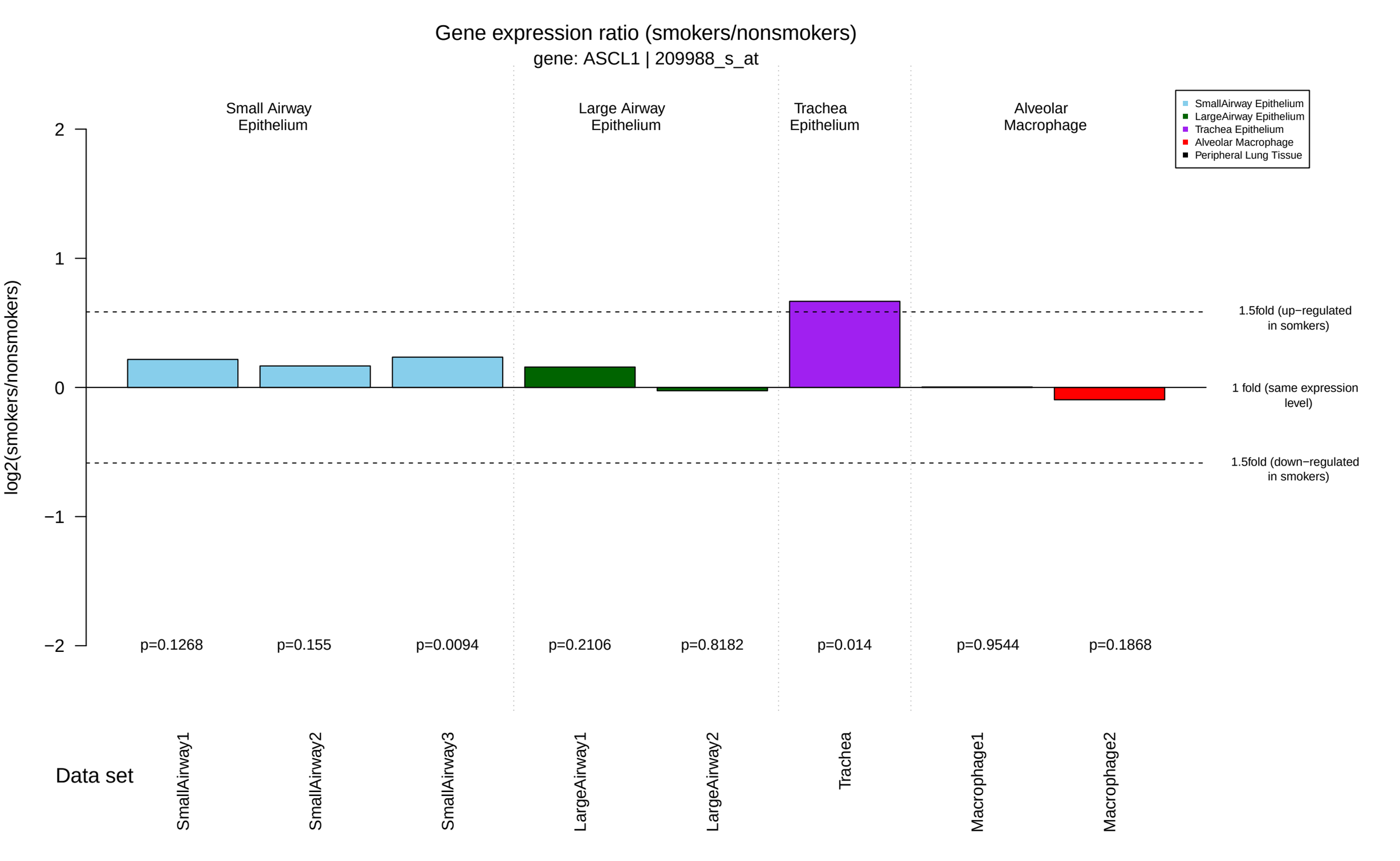
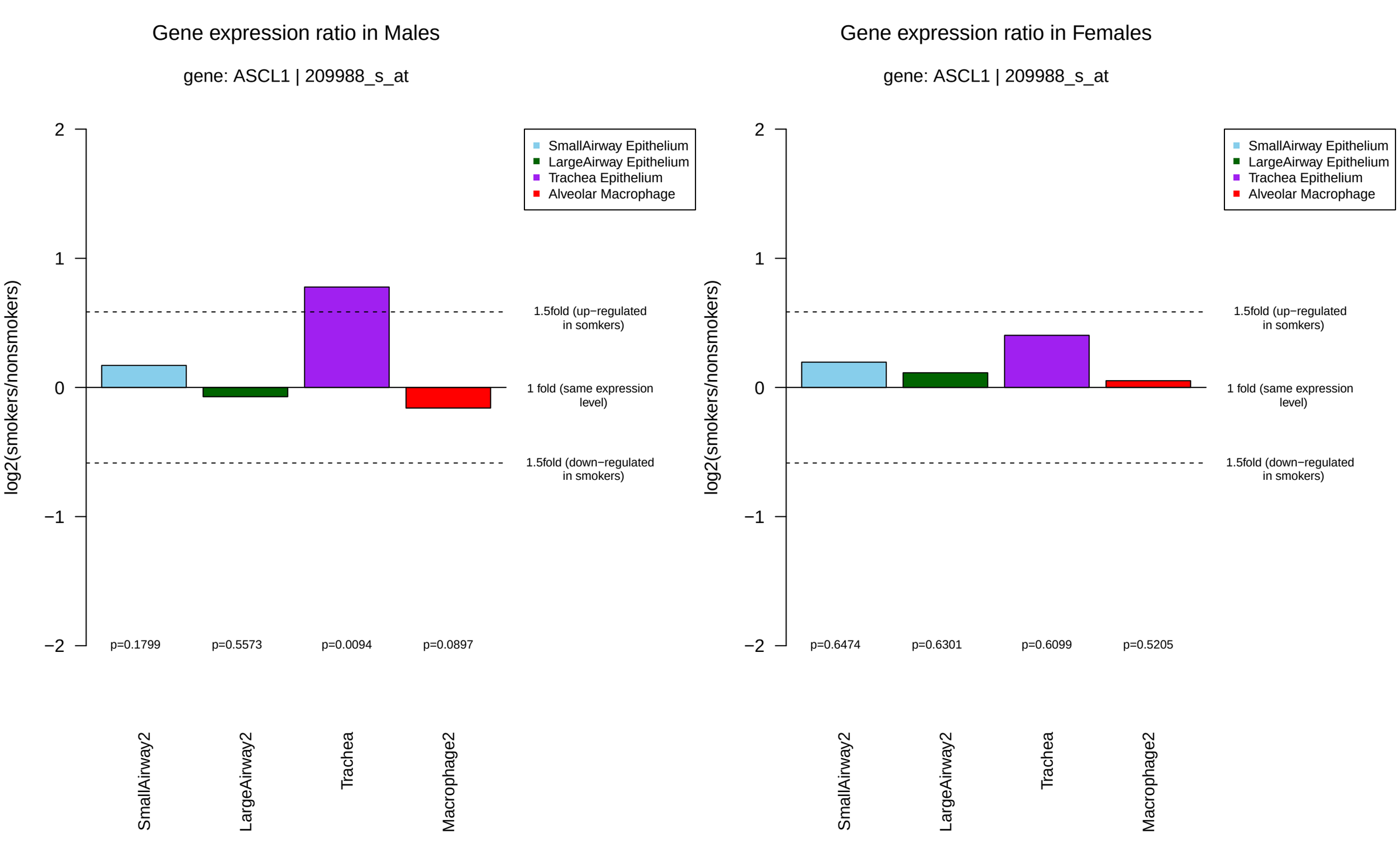
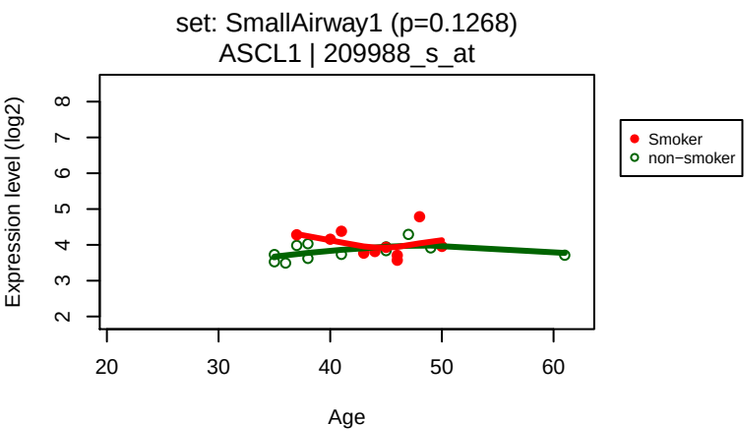
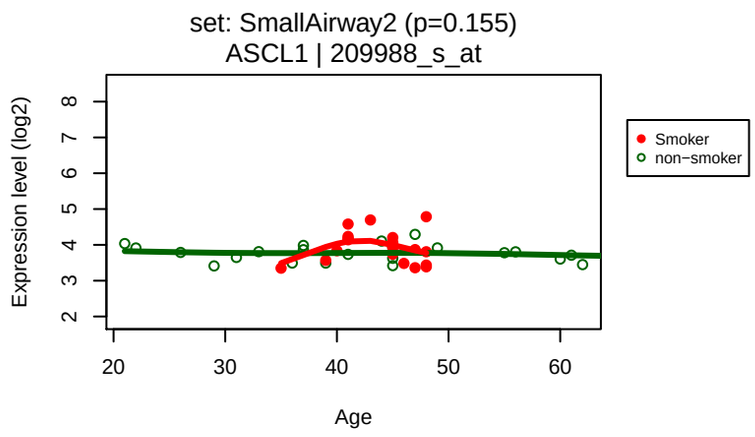
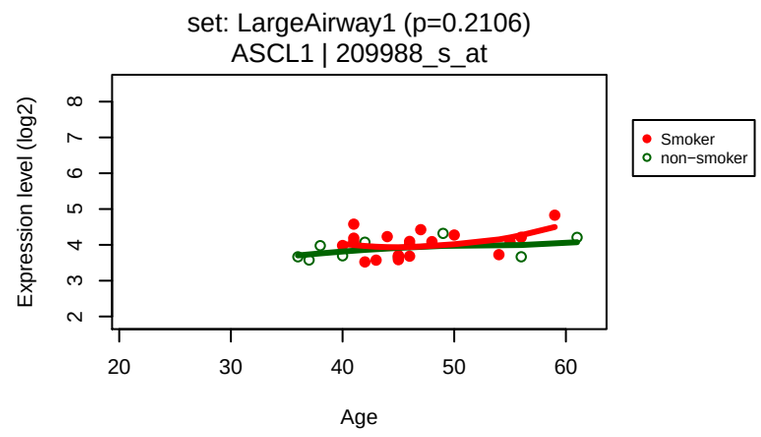

Gene expression of smokers and nonsmokers
gene: ASCL1 | 209988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1268
p=0.155
p=0.0094
p=0.2106
p=0.8182
p=0.014
p=0.9544
p=0.1868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

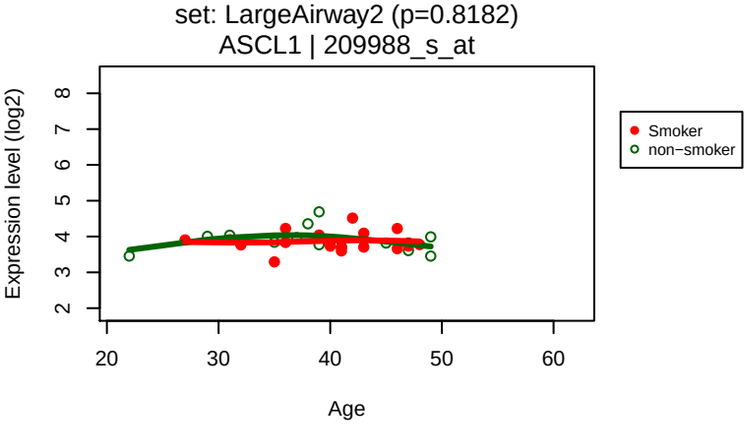

Gene expression ratio (smokers/nonsmokers)
gene: ASCL1 | 209988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1268
p=0.155
p=0.0094
p=0.2106
p=0.8182
p=0.014
p=0.9544
p=0.1868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

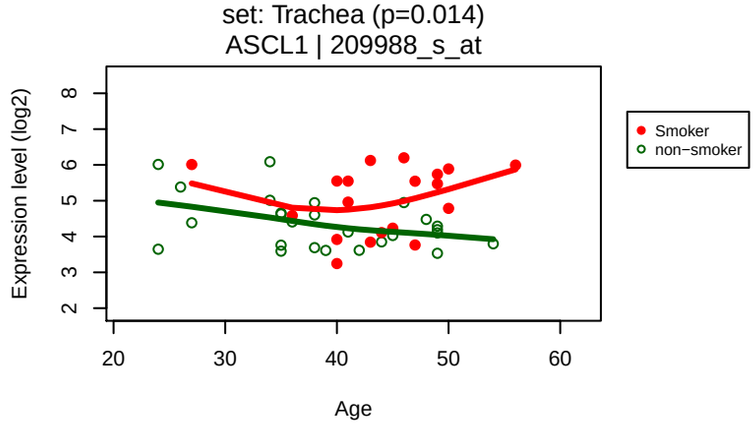

Gene expression ratio in Males
gene: ASCL1 | 209988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1799
p=0.5573
p=0.0094
p=0.0897
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASCL1 | 209988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6474
p=0.6301
p=0.6099
p=0.5205
SmallAirway2
LargeAirway2
Trachea
Macrophage2

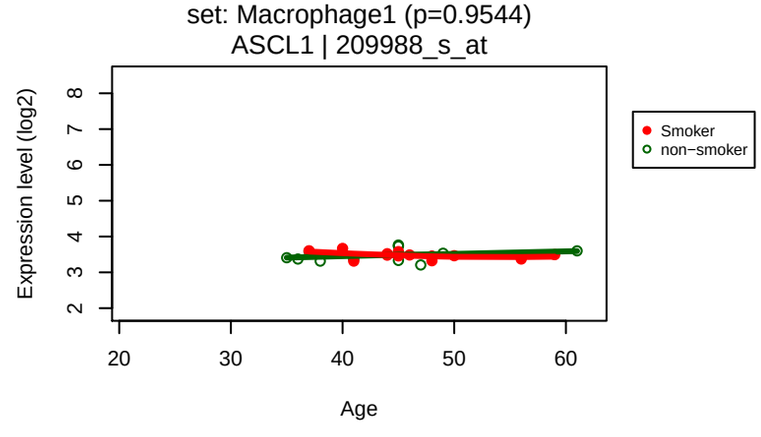

set: SmallAirway1 (p=0.1268)
ASCL1 | 209988_s_at
Expression level (log2)
Age
Smoker
non−smoker

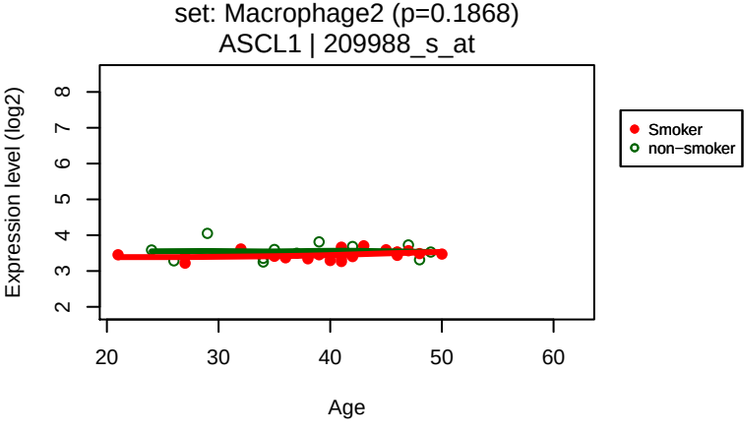

set: SmallAirway2 (p=0.155)
ASCL1 | 209988_s_at
Expression level (log2)
Age
Smoker
non−smoker

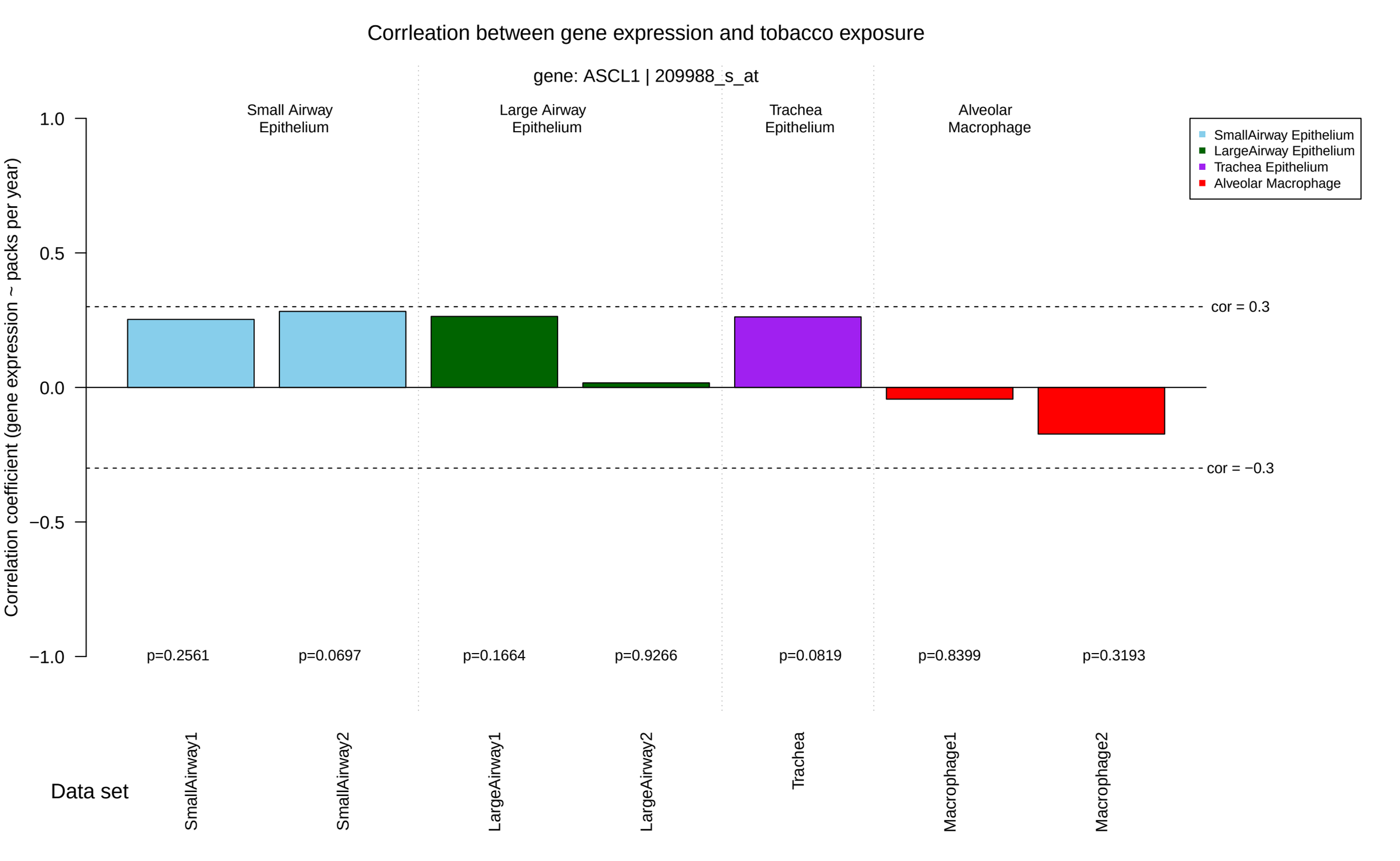

set: LargeAirway1 (p=0.2106)
ASCL1 | 209988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8182)
ASCL1 | 209988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.014)
ASCL1 | 209988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9544)
ASCL1 | 209988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1868)
ASCL1 | 209988_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASCL1 | 209988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2561
p=0.0697
p=0.1664
p=0.9266
p=0.0819
p=0.8399
p=0.3193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2